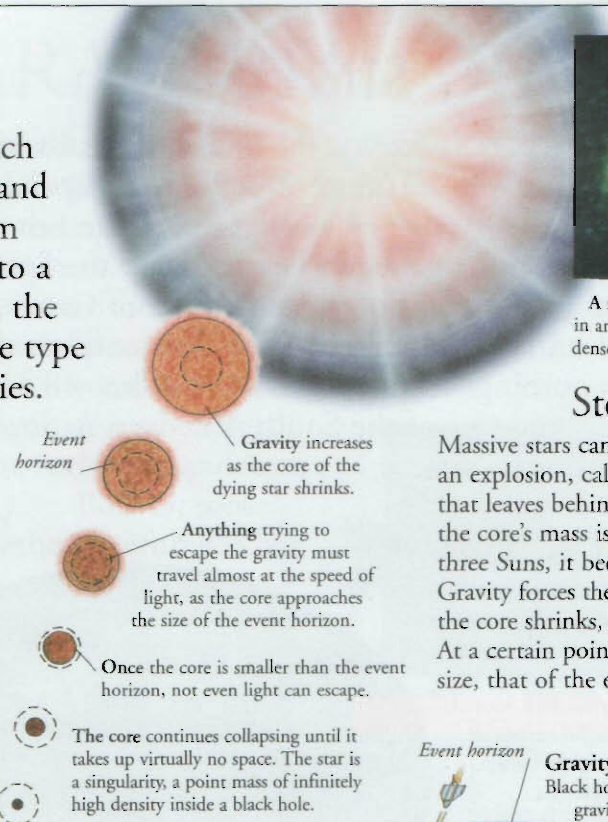
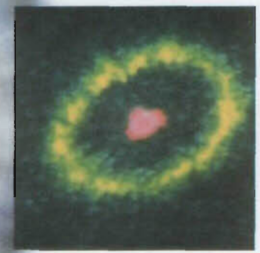
BLACK HOLES

ASTRONOMERS HAVE SPENT much time analysing how stars form and how they develop. One problem was to explain what happened to a massive star at the end of its life. In 1967, the term "black hole" was used to describe one type of object that is left when a massive star dies. Four years later, Cygnus X-1 was found, the first candidate for a black hole.

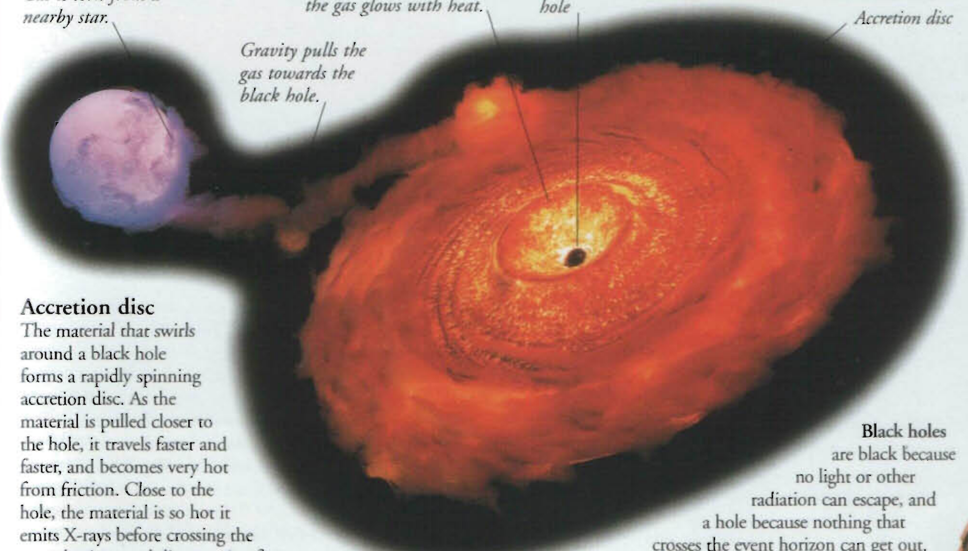




A massive star ends its days in an explosion, leaving a very dense core that then collapses.

Detecting a black hole

Black holes appear black because nothing, not even light, can escape from their powerful gravity. Astronomers cannot detect them directly, but can "see" them because of the effect their gravity has on everything around them, such as gas from a nearby star. The boundary of the black hole is called the event horizon. Material pulled in towards the hole is swirled around by the gravity, forming a disc, before crossing the horizon.

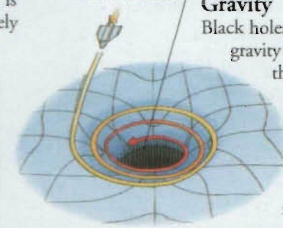


Accretion disc
The material that swirls around a black hole forms a rapidly spinning accretion disc. As the material is pulled closer to the hole, it travels faster and faster, and becomes very hot from friction. Close to the hole, the material is so hot it emits X-rays before crossing the event horizon and disappearing forever.

Black holes are black because no light or other radiation can escape, and a hole because nothing that crosses the event horizon can get out.

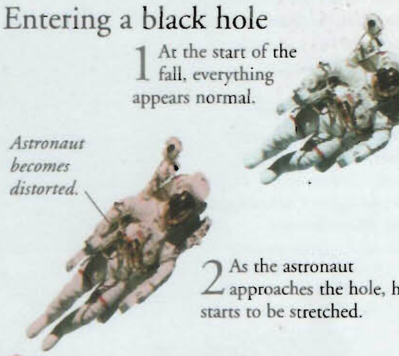
Stellar collapse

Massive stars can end their lives in an explosion, called a supernova, that leaves behind a central core. If the core's mass is more than that of three Suns, it becomes a black hole. Gravity forces the core to collapse. As the core shrinks, its gravity increases. At a certain point it reaches a critical size, that of the event horizon.



Gravity
Black holes have incredibly strong gravity which pulls in anything that comes close enough. Anything pulled in beyond the event horizon will be squashed to near infinite density and never escapes.

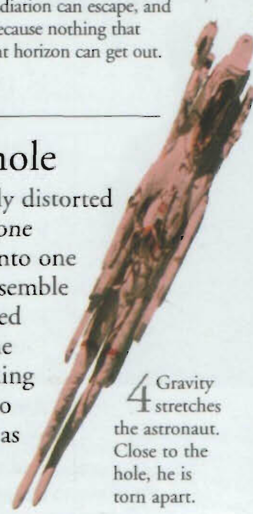
Entering a black hole



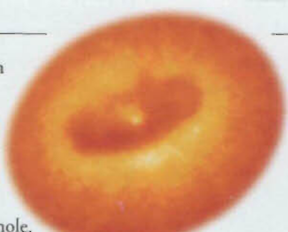
1 At the start of the fall, everything appears normal.

2 As the astronaut approaches the hole, he starts to be stretched.

3 Light is also stretched to a longer wavelength so the astronaut appears redder.



4 Gravity stretches the astronaut. Close to the hole, he is torn apart.



Galaxy NGC 4261 in the constellation of Virgo has what appears to be a huge accretion disc – 30 million light years across – swirling around a huge black hole.

Supermassive holes

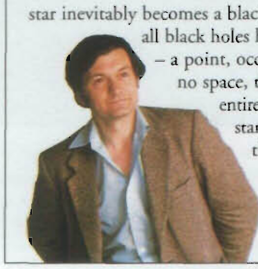
Some galaxies have very active centres that give out large amounts of energy. An object of powerful gravity, such as a supermassive black hole, could be the cause of the activity. Such a hole would be a hundred million times more massive than the Sun.

Inside a black hole

Space and time are highly distorted inside a black hole. Anyone unlucky enough to fall into one would be stretched to resemble spaghetti, as gravity pulled more on the feet than the head. An observer watching the person fall would also see time running slower as the person fell towards the event horizon.

Roger Penrose

The English mathematician Roger Penrose (b. 1931) theorizes on the nature of space and time. He has shown that a massive collapsing star inevitably becomes a black hole, and that all black holes have a singularity – a point, occupying virtually no space, that contains the entire mass of the dead star. Penrose believes the singularity is always hidden by an event horizon.



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